

Capstone Project: Spotify Music Recommendation System

Executive Summary:

This project develops and evaluates multiple recommendation system models to improve content relevancy and increase user engagement on a music streaming platform. The dataset captures interactions between users and songs; specifically play counts; providing the foundation for training models that predict what a user is likely to listen to next.

Recommendation systems are essential to any interaction-driven platform, whether in e-commerce, streaming, or social media. In the music streaming context, recommending the right song at the right time directly impacts user attraction and satisfaction. Given the dataset's characteristics "Post Cut-off" 3,155 users, 563 songs, and 117,876 observed interactions — five recommendation approaches were implemented and evaluated: rank-based filtering, user-user collaborative filtering, item-item collaborative filtering, matrix factorization (SVD), Coclustering and content-based filtering.

Each model was assessed on Precision@K, Recall@K, F1 Score, and RMSE, with GridSearchCV applied to optimize hyperparameters across all collaborative and matrix factorization models. The optimized SVD model emerged as the strongest performer, achieving the highest Recall **(0.661)** and F1 Score **(0.502)** among all models — the metrics most relevant to recommendation quality. Its computational efficiency, completing training in under a minute compared to several minutes for KNN-based approaches, further supports its suitability for production deployment at scale.

The proposed final architecture combines SVD as the primary personalization engine, supported by a rank-based fallback for new users solving the cold start problem and content-based filtering for new songs — ensuring every user receives relevant recommendations regardless of their interaction history.

Key Findings:

- All five models were successfully implemented and evaluated on a consistent rating scale after applying a $\text{play_count} \leq 5$ filter, which resolved the rating scale mismatch and produced reliable RMSE values in the range of **1.01–1.09**
- SVD optimized outperformed all other models on Recall **(0.661)** and F1 **(0.502)** — the two most critical metrics for surfacing relevant songs in the top-K recommendations
- KNN-based models required significantly longer training times than SVD, making them impractical at production scale despite competitive accuracy on individual predictions

- CoClustering showed no improvement after hyperparameter tuning, confirming it is not well suited to this dataset's size and interaction density
- Content-based filtering successfully grouped songs by artist similarity, validating TF-IDF on metadata as a viable cold-start fallback for new songs with no interaction history

Final Model Specifications — SVD Optimized:

Parameter	Value
Algorithm	Singular Value Decomposition (SVD)
Library	scikit-surprise
n_epochs	30
lr_all (learning rate)	0.001
reg_all (regularization)	0.1
Rating scale	0 – 5
Train/test split	80% / 20%
Evaluation K	30
Relevance threshold	1.5
Tuning method	GridSearchCV
RMSE	1.0214
Precision@30	0.405
Recall@30	0.661

F1@30	0.502
Training time	< 1 minute

Problem Summary:

As modern life becomes increasingly fast-paced, the time available to discover new artists and songs has significantly diminished. At the same time, technology has transformed how people consume art and entertainment — and for any internet-based platform, the ability to recommend relevant content is no longer a differentiator but a necessity. User engagement, session length, and ultimately revenue is all directly tied to how well a platform can surface the right content at the right moment.

Spotify is one such audio content provider with a huge market base across the world. With the daily increase of volume of songs becoming available on the internet, searching for songs of interest has become a tedious task in itself. However, Spotify has grown significantly in the market because of its ability to recommend the ‘best’ next song to each and every customer based on a huge preference database gathered overtime-millions of customers and billions of songs. This is done by using smart recommendation systems that can recommend songs based on users’ likes.

This project tackles the same core problem: given a dataset of user-song interactions captured through play counts, how can we build a recommendation system that accurately predicts what a user wants to listen to next? Solving this problem has direct business implications — better recommendations mean longer listening sessions, higher retention rates, and stronger platform loyalty.

Dataset Description:

The dataset is sourced from the **Million Song Dataset**, widely used in music information retrieval research. It consists of two files joined on a common song identifier:

- **count_data** — captures user-song interactions with three fields: `user_id`, `song_id`, and `play_count` (number of times a user played a song). This forms the core interaction matrix for all **collaborative filtering and matrix factorization models**.

- **song_data** — provides song metadata including title, release, artist_name, and year, used for exploratory analysis and the content-based recommendation model.

Raw Data: The raw combined dataset contains **1,000,000 interactions** across **76,353 unique users** and **10,000 unique songs** before any filtering is applied.

Cutoff & Filtering: To reduce matrix sparsity and ensure every user and song has sufficient interaction history for meaningful model training, three sequential filters were applied:

- Users with fewer than **90 song interactions** were removed
- Songs interacted with by fewer than **120 users** were removed
- Records with a **play count greater than 5** were removed to align with the rating scale and eliminate extreme outliers

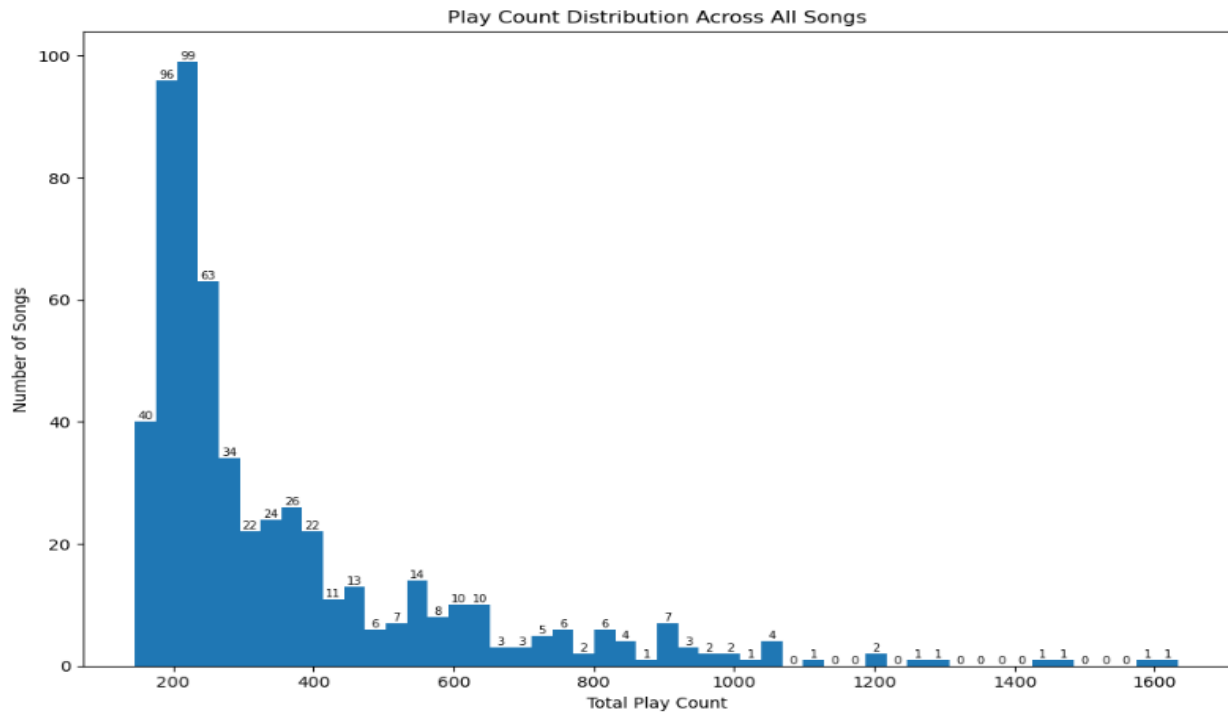
The observed interaction rate of just **6.5%** out of ~1.8 million possible user-song pairs in the filtered dataset highlights the core challenge — extreme sparsity — which directly shaped the choice of models. User and song IDs were label-encoded to numeric format for model compatibility, and play count serves as an implicit proxy for user preference, with the key limitation that it captures listens but not negative signals such as skips or partial plays.

EDA Key Findings:

Exploratory analysis of the filtered dataset revealed five key patterns that directly shaped the modelling approach:

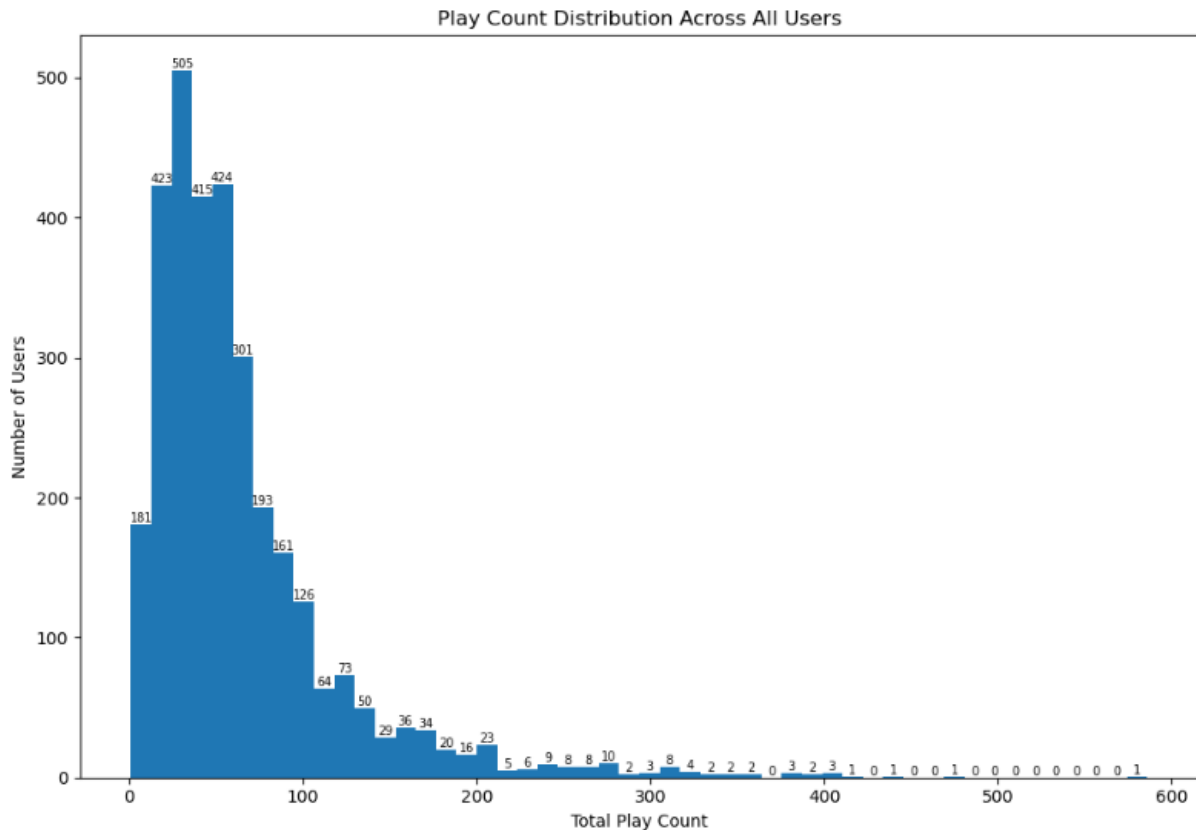
Play Count Distribution — The distribution is heavily right-skewed with **61.5% of interactions being single plays**, confirming that most listening behaviour is exploratory rather than deeply engaged. This sparsity motivated the use of a relevance threshold of 1.5 in model evaluation.

Figure:1



User Engagement — Interaction counts range from 1 to 243 songs per user, with most users falling in the **10–40 song range** — sufficient history for collaborative filtering to compute meaningful similarity, though lower-end users remain a cold-start concern.

Figure 2:

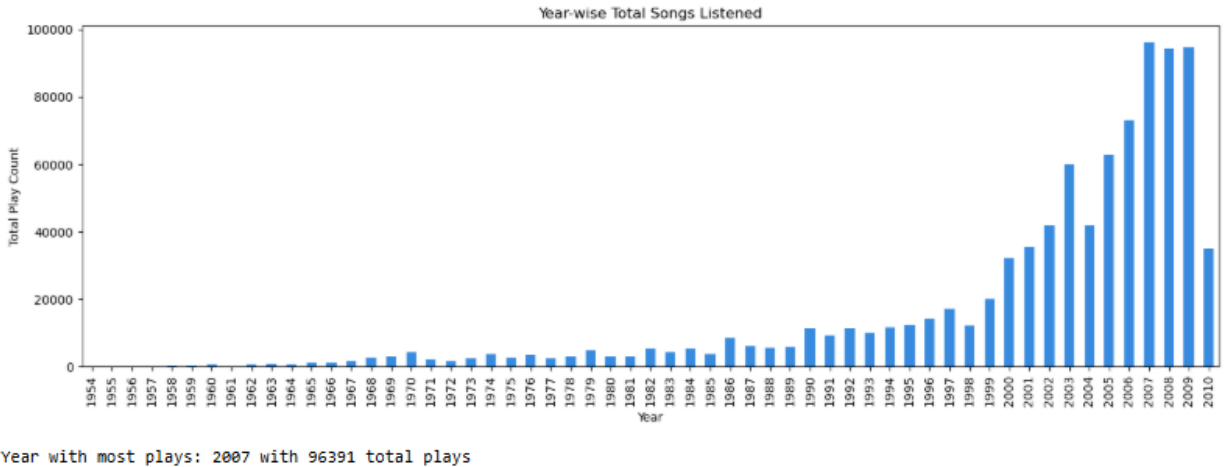


Song Popularity — Popularity is highly concentrated, with the top song recording **751 unique user interactions** while most songs sit near the 120-interaction cutoff. This long-tail distribution directly motivates the rank-based model as a popularity baseline.

Artist Concentration — **Coldplay leads with 17,595 cumulative plays** across 3,375 unique artists, followed by The Black Keys and Kings of Leon. This strong artist-level clustering validated artist name as the primary feature in the content-based model.

Year-wise Trends — **2007 recorded the highest total plays at 96,391**, reflecting a recency bias toward mid-to-late 2000s music coinciding with the rise of digital streaming.

Figure 3:



The findings from exploratory analysis were not treated in isolation — each pattern directly informed a specific modelling decision, as summarised below:

EDA Finding	Modelling Decision	Why
61.5% single-play interactions	Rank-based fallback for cold-start users	Weak signals mean new or inactive users lack sufficient history for personalized models
10–40 songs per user	User-User and Item-Item KNN included	Moderate interaction density is sufficient for neighborhood-based similarity
Only 563 unique songs	SVD prioritized over Item-Item	A small catalog limits co-interaction overlap between songs, making item-to-item similarity weaker — SVD's latent factors generalize better across sparse pairs
Strong artist clustering (Coldplay 17,595 plays)	Artist name as primary content-based feature	The dominance of specific artists confirms that artist identity is the strongest similarity signal available in the metadata
2007 peak listening year	Recency bias acknowledged in rank-based model	The dataset skews toward mid-2000s releases, meaning popularity-based recommendations will naturally favour this era

Solution Design:

A number of recommendation models and methods were explored as part of the solution design, including ranking rank-based filtering, user-user collaborative filtering, item-item collaborative filtering, matrix factorization (SVD), Cluster-based and content-based filtering.

Rank-Based Filtering — recommends globally popular songs as a cold-start baseline for new users with no interaction history.

User-User Collaborative Filtering — identifies users with similar listening patterns and recommends what those neighbors enjoyed. Optimized with pearson_baseline similarity, **k=40, min_k=6 via GridSearchCV, achieving Recall 0.641 and F1 0.505.**

Item-Item Collaborative Filtering — finds songs similar to those already played based on co-interaction patterns. Achieved the **lowest RMSE (1.0117)** but weaker Recall (**0.583**), confirming that with only 563 songs, item-to-item overlap is too limited for strong similarity signals.

Matrix Factorization (SVD) — learns compact latent factor representations from the interaction matrix without requiring direct user or item overlap. Uniquely suited to the dataset's **6.5%** sparsity. Optimized with **n_epochs=30, lr_all=0.001, reg_all=0.1, yielding the best Recall (0.661) and F1 (0.502).**

CoClustering — groups users and songs into clusters for recommendation. Despite GridSearchCV tuning, produced identical metrics before and after optimization, confirming it is not well suited to this dataset.

Content-Based Filtering — recommends songs using TF-IDF cosine similarity across artist, title, and album metadata. Requires no interaction history, making it the natural fallback for new songs. Successfully surfaced Foo Fighters tracks for "Learn To Fly", validating the artist-level similarity signal.

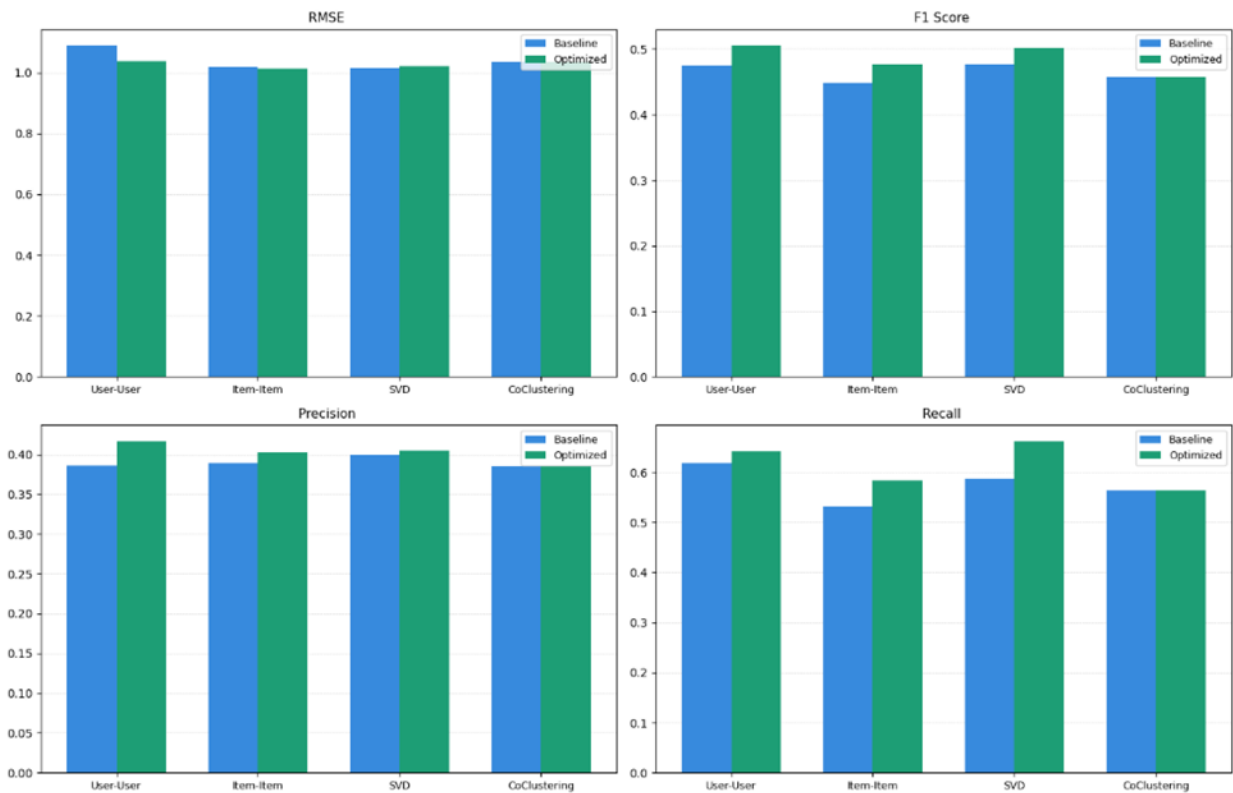
Model	Key Parameters	RMSE	Precision	Recall	F1
KNN User-User (baseline)	cosine, user_based=True	1.0875	0.386	0.619	0.475
KNN User-User (optimized)	pearson_baseline, k=40, min_k=6	1.0365	0.416	0.641	0.505
KNN Item-Item	pearson_baseline,	1.0188	0.389	0.532	0.449

(baseline)	user_based=False				
KNN Item-Item (optimized)	pearson_baseline, k=40, min_k=3	1.0117	0.403	0.583	0.477
SVD (baseline)	default parameters	1.0152	0.4	0.587	0.476
SVD (optimized)	n_epochs=30, lr_all=0.001, reg_all=0.1	1.0214	0.405	0.661	0.502
CoClustering (baseline)	default parameters	1.0344	0.385	0.563	0.457
CoClustering (optimized)	n_cltr_u=7, n_cltr_j=3, n_epochs=30	1.0344	0.385	0.563	0.457
Content-Based	TF-IDF cosine, artist+title+album	—	—	—	—
Rank-Based	avg play count, min 100 interactions	—	—	—	—

Using the data set with cutoff of the user, song, and listens. The final proposed solution is the **optimized matrix factorization**, which has been optimized with GridSearchCV to optimize hyperparameters. Using a cutoff of the user, song, play count fields was essential to better make the prediction process meaningful.

Figure 4 presents a four-panel comparison of RMSE, F1 Score, Precision, and Recall across all collaborative filtering models — User-User, Item-Item, SVD, and CoClustering — with baseline and optimized variants shown side by side. The filter of `play_count ≤ 5` was applied consistently across all models to ensure a fair and meaningful comparison on a standardized rating scale.

Model Performance Comparison — Baseline vs Optimized (play_count ≤ 5 filter applied)



Several key observations stand out. First, all models perform within a remarkably tight RMSE band of **1.01–1.09**, confirming that the play count filter successfully eliminated the rating scale mismatch seen in earlier runs and produced reliable, comparable predictions across the board. Second, optimization consistently improved performance across all four metrics for every model, validating the use of GridSearchCV as a tuning strategy. Third and most critically, the SVD optimized model achieves the highest F1 Score (**0.502**) and the highest Recall (**0.661**) — the two metrics that matter most for a recommendation system where the primary goal is to surface as many relevant songs as possible within the top-K results. CoClustering was the only model that showed no improvement after tuning, with baseline and optimized results remaining identical, indicating the algorithm has plateaued on this dataset.

Key Limitations

While the models demonstrate strong performance within the scope of this dataset, three limitations are worth acknowledging:

- **Weak interaction signals** — play count alone is a limited factor for preference. It captures listens but not negative signals such as skips, partial plays, or repeated listens within a single session. Richer implicit feedback would produce more accurate models.
- **Thin content features** — the content-based model relies solely on text metadata (artist, title, album). The absence of audio features such as genre, tempo, and energy limits its ability to capture true musical similarity beyond artist-level grouping.
- **Dataset scale** — with only 563 songs and 3,155 users after filtering, the models are trained on a relatively small interaction space. Performance on a real-world catalog of millions of songs and users may differ, particularly for Item-Item and CoClustering which showed the weakest scalability characteristics.

Recommendations for Implementation:

Data Engineering “pipelines”:

- Implement real-time logging to capture richer signals — skips, partial plays, saves, and repeat listens — beyond raw play counts
- Build an automated weekly retraining pipeline with RMSE-based drift monitoring
- Pre-compute SVD latent factor vectors in Redis for real-time serving at scale

Product & UX:

- Deploy the three-tier hybrid architecture with automatic fallback routing
- Introduce diversity constraints to prevent over-concentration on mainstream artists
- Surface explainability alongside recommendations — “Because you listened to Foo Fighters”

Business Leadership:

- Run a structured A/B test — SVD to 50% of users, rank-based to 50% — for a minimum of two weeks measuring click-through rate, listen-through rate, and 7-day retention
- Set deployment thresholds: $\text{Recall}@30 \geq 0.70$ and 7-day retention uplift $\geq 5\%$

- Allocate budget for Spotify Audio Features API integration — the highest-impact single improvement available

Business Impact & Deployment

Implementing the SVD-based recommendation system is projected to deliver measurable business value across three dimensions:

- **User engagement** — personalized recommendations reduce the time users spend searching for music, increasing session length and daily active usage
- **Retention/Attraction** — users who consistently discover relevant content are significantly less likely to churn to competing platforms
- **Catalog utilization** — a well-tuned recommendation system surfaces long-tail content beyond the top 50 popular songs, increasing overall catalog engagement and benefiting independent artists on the platform

Deployment Architecture SVD latent factors are pre-computed offline and stored in a low-latency key-value store such as Redis. At request time, the system retrieves the user's factor vector and computes similarity scores against all song vectors in milliseconds — making real-time recommendations feasible at scale.

Hybrid Fallback Strategy A three-tier serving architecture ensures every user receives a recommendation regardless of their interaction history:

New user (no history) - **Rank-Based**

New song (no play data) - **Content-Based**

Active user - **SVD Optimized**

Monitoring & A/B Testing The system should be validated in production through a structured A/B test — deploying SVD to 50% of users and rank-based to the remaining 50% — measuring click-through rate, listen-through rate, and 7-day retention as primary success metrics. Model performance should be monitored daily to weekly, with retraining triggered if RMSE increases by more than 10% week-over-week or if recommendation diversity drops below acceptable thresholds.

Expected Benefits & Costs

	Detail
Benefit	Increased session length — less time searching, more time listening
Benefit	Reduced churn — consistent relevancy drives platform retention
Benefit	Improved catalog utilization — long-tail and independent artists surface
Benefit	Scalable — SVD retrains in under a minute at low computational cost
Cost	Redis infrastructure for real-time vector serving
Cost	4–6 weeks engineering effort for retraining pipeline
Cost	Minimum 2-week A/B test before statistically significant results

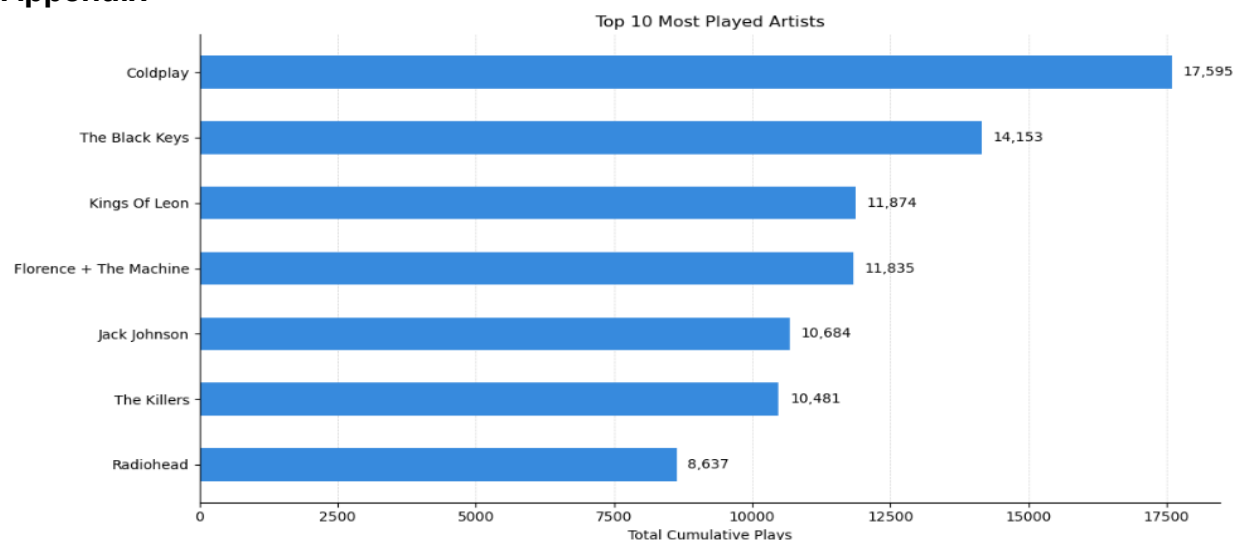
Key Risks & Challenges

- **Cold-start** — new users and songs receive generic recommendations until sufficient interaction history accumulates. Mitigated by rank-based and content-based fallbacks but not eliminated. Onboarding prompts such as genre preference selection at signup can accelerate the transition to personalized recommendations
- **Popularity bias** — SVD and rank-based models both tend to over-recommend mainstream content. Without diversity constraints the system risks creating a feedback loop where dominant artists receive disproportionate exposure, disadvantaging independent artists
- **Data drift** — the dataset skews heavily toward 2007-era music. User tastes evolve continuously and a model without regular retraining will progressively diverge from current listening trends
- **Scalability gap** — the model was trained on 563 songs and 3,155 users. Performance at real-world scale of millions of songs and users requires load testing and architectural validation before full deployment

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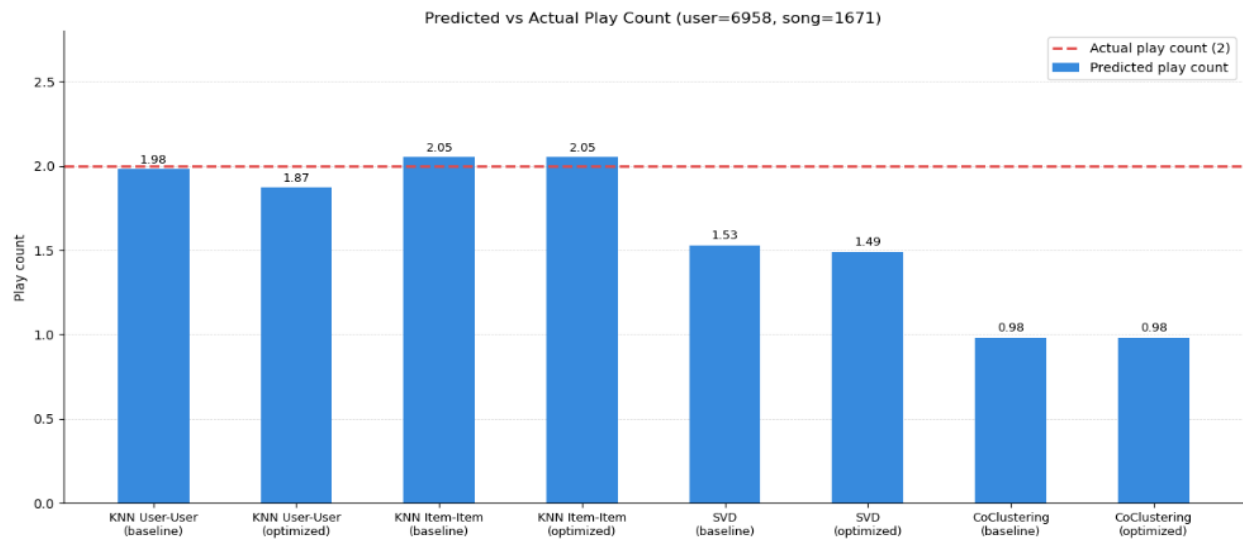
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Appendix



The chart confirms the strong artist-level concentration observed in the dataset — the top 7 artists collectively account for over 85,000 cumulative plays out of 200,396 total, representing approximately **42% of all listening activity**. This dominance of mainstream acts directly validated using artist name as the primary feature in the

content-based filtering model, and highlights the popularity bias risk that diversity constraints must address in production deployment.



The chart compares each model's predicted play count for the sample user-song pair (user=6958, song=1671) against the actual value of 2.00. With the **play_count ≤ 5** filter applied, all models now predict within a reasonable range — a significant improvement over the unfiltered results where SVD baseline predicted the ceiling value of 5.00 for every song. KNN Item-Item produces the closest prediction at 2.05, while CoClustering underestimates most significantly at 0.98, consistent with its weaker overall metrics across all evaluation criteria.